

# Stephen Welch

Gainesville, VA

☎ (+1) 703-639-7926 | ✉ [stephenwelchva@vt.edu](mailto:stephenwelchva@vt.edu) | 📄 [github.com/StephenWelch](https://github.com/StephenWelch) | 🔗 [linkedin.com/in/stephenwelchva](https://www.linkedin.com/in/stephenwelchva)

## Education

### M.S. in Computer Engineering: Software & Machine Intelligence

May 2025 | GPA: 3.88

Relevant Courses: Learning Theory for Dynamics & Control | Machine Learning | Adaptive Control

Virginia Polytechnic Institute

Aug. 2023 - May 2025

### B.S. in Computer Engineering: Controls, Robotics & Autonomy

Minor in Computer Science | GPA: 3.27

Virginia Polytechnic Institute

Aug. 2019 - May 2023

## Skills

- Tools
- Git, Gradle, JetBrains IDEs, Visual Studio, GNUPlot,  $\LaTeX$
- Frontend
- JavaFX, Swing, Qt, ASP.NET
- Languages
- Python, Java 8, C++, C#, SQL

## Work Experience

### Software Engineer - Humanoid Controls

Apptronik

Austin, TX

Jun. 2024 - Present

- Built training, evaluation & inference pipeline to deploy the first reinforcement learning policies on the Apollo humanoid robot using **Isaac Lab** & **MuJoCo**
- Developed policies and sim-to-real pipelines for whole-body teleoperation, skill imitation from motion capture, and natural lo-

### Robotics Researcher

Terrestrial Robotics Engineering and Controls Lab

Blacksburg, VA

Jan. 2020 - Present

- Researched applications of convex optimization and reinforcement learning to robotics tasks
- Designed **Java** software enabling QP-based whole-body control of a 32 DoF parallel-actuated bipedal robot for balancing & lo-

### Research Engineer

Occupational Ergonomics & Biomechanics Lab

Blacksburg, VA

Jan. 2024 - Dec. 2024

### Machine Learning Intern

Shield AI

Alexandria, VA

Dec. 2020-May 2023

- Applied model predictive control with learned dynamics models to autonomous jets, enabling a 90% improvement in sample efficiency
- Developed optimization-based approaches to multi-vehicle coordination & target selection
- Built air-to-air combat simulation framework for large-scale (4v4+) dogfights in **PyTorch** & **RLlib** used across multiple govern-

### Enterprise Applications Intern

Serco North America

Herndon, VA

Mar. 2020 - Aug. 2020

- Wrote utilities to verify website integrity using **Java**, **JavaFX**, **Swing**, and **Selenium** with interactive graph visualizations
- Performed bugfixes and quality-of-life changes on widely used TA/NDA request web application built using **C#**, **ASP.NET** and a **SQL** database
- Eliminated time-consuming manual labor for Contracts department using **PowerShell** scripts to automatically update macro-enabled spreadsheets

## Projects

### AUV Fault Detection

Center for Marine Autonomy and Robotics

2023

- Integrates with IMU-based state estimator to detect actuator & sensor faults on the 690-type submersible robot, increasing reliability during long-term surveys of the oceanbed
- Implemented & validated in **C++** **ROS**-based simulation

### Miniature Bipedal Walker

Personal Project

In Progress

- Designing and building torque-controlled 3 degree-of-freedom parallel-actuator bipedal robot as a <\$1000 learning platform for motion planning & control

## Multi-Agent Heron Manager

Heron Systems

2021

- **Python** framework enabling multi-agent interaction with multi-threaded and distributed compute capability
- Designed in collaboration with senior engineers and developed with 2 fellow interns
- Integrates with open-source and in-house reinforcement learning libraries
- Implements multi-agent air-to-air combat environment using JSBSim and RLLib

## Robot Arm

Actively maintained in-house

Personal Project

2019

- Designed and 3D printed low-cost 2 degree-of-freedom robot arm using SOLIDWORKS
- Wrote **C++** Arduino code receive commands from PC over serial and drive stepper motors
- Created **Qt** graphical interface in **C++** to control arm over serial
- Implemented inverse kinematics algorithm for arm

## Accomplishments

---

2018-2019 **Advanced Computer Studies Student of the Year**, Battlefield High School

Haymarket, VA

2018-2019 **Dean's List**, Northern Virginia Community College

Haymarket, VA

2017-2019 **National Semi-Finalist**, CyberPatriot

Haymarket, VA

Apr. 2018 **Presenter for Effective Student Leadership in the FIRST Robotics Competition**, FRC World Championship

Detroit, MI

Apr. 2022 **Author for A Mapping Approach to Achieve Torque Control for Parallel-Actuated Robotic Systems**, ASME 2022

TBD